

Nitish Shukla, PhD CS (2nd Year)

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Links: Portfolio: nitishshukla86.github.io, LinkedIn: linkedin.com, Google Scholar: scholar.google.com, Github: github.com.

Employment history

Adobe Research, PhD Intern, May 2025 - Aug 2025

Attribute-based style validation via MLLM induced embeddings to enable fine-grained analysis and retrieval of artistic styles.
Attribute based image splitting achieves 2-3% increase in recall. [Under review at CVPR 2026]

Micron Technology, Data Scientist II , Aug 2022 - Aug 2023

Developed a synthetic data generation pipeline to finetune segmentation pipeline (proprietary) for defective die detection.
Synthetic data increased defect shape mIoU by 9%.[\[Paper\]](#).

Next Education, R&D Engineer, May 2019 - Aug 2020

Ongoing Projects

• Desired Difficulty for Data-Efficient Multi-Image Preference Optimization

“Can multi-image preference optimization be done using less annotated samples?” We propose to exploit the concept of desirable difficulties to generate multi-image data samples by leveraging intrinsic relationships between images through carefully designed prompts which previous works ignore. Our data creation pipeline reduced annotation requirements by ~60% while achieving 5–6% gains across multi-image benchmarks. *(Target: ACL 2026)*

• Style-Aware Open-Set Retrieval via MLLMs

“MLLMs can perform MCQ based image-retrieval, but can we use them to perform open-set retrieval?” We 1) propose a framework exploit MLLM zero-shot understanding to produce rich representation for image-text pair and 2) employ our extraction method to do attribute-aware art-style retrieval. Our proposed method disentangles attributes such as brushstrokes, color palette, etc. form an art image and performs attribute aware retrieval and achieves 2–3% mAP without needing any training. *(Under review)*

• MLLMs are Prosopagnic

Designed a benchmark, large-scale multi-image data generation pipeline, and fine-tuning regime to address MLLM weaknesses in facial understanding. Achieved 6–8% accuracy improvement on multi-image face understanding tasks. *(Target: ACL 2026)*

Research Publications

1. **Nitish Shukla**, Arun Ross, “Facial Demorphing from a Single Morph Using a Latent Conditional GAN”, In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2025 [**IAPR Best Student Paper Award**].
2. **Nitish Shukla**, Arun Ross, “diffDemorph: Extending Reference-Free Demorphing to Unseen Faces”, In Proceedings of IEEE International Conference on Image Processing (ICIP), 2025.
3. **Nitish Shukla**, Arun Ross, “dc-GAN: Dual-Conditioned GAN for Face Demorphing From a Single Morph”, In Proceedings of IEEE International Conference on Automatic Face and Gesture Recognition (FG), 2025.
4. **Nitish Shukla**, Arun Ross, “Metric for Evaluating Performance of Reference-Free Demorphing Methods”, In Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), 2025.
5. **Nitish Shukla**, Arun Ross, “Face Demorphing via Identity Preserving Image Decomposition”, In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2024.
6. **Nitish Shukla**, Sudipta Banerjee, “Generating Adversarial Attacks in Latent Space”, In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023.
7. **Nitish Shukla**, “SDeMorph : Towards Better Component Face Extraction from Single Morph”, In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2023.

Education

PhD in Computer Science, Jan 2024 - Present

Michigan State University, MI, USA

Master of Science in Computer Science, Dec 2020 - Jun 2022

Chennai Mathematical Institute, Chennai, India

Master of Science in Mathematics, Jul 2017 - Jun 2019

Indian Institute of Technology (IIT), Guwahati, Guwahati, India

Skills

Face Biometrics (*Experienced*), Deep Learning (*Experienced*), Computer Vision (*Experienced*), Adversarial ML (*Skillful*), Bayesian Theory, Graph Theory, Mathematics, PyTorch, Scikit-learn, OpenCV, MLFlow, Pandas, LLM Planning (*Skillful*).

References

Arun Ross, Michigan State University (rossarun@msu.edu), K.V. Subramanayam, CMI (kv@cmi.ac.in, +91 9445 390 492),
Sudipta Banerjee, IIIT-H (sudipta.b@iiit.ac.in).